Lesson 2: Turtle Graphics

Module 1 | 60 minutes | turtle, loops, functions | Grades 7–11 · ages 12–16

Name: _____ Class: _____ Date: _____

Practise what you learned today. Read each task carefully.

1. Key words

Match each word to what it means. Write the letter.

a. turtle ___ A. How far to turn, measured in degrees; a full turn is 360.
b. forward(n) ___ B. Turns the turtle by a degrees.
c. right(a) / left(a) ___ C. Moves the turtle n steps, drawing a line if the pen is down.
d. angle ___ D. A drawing robot in Python that moves and draws lines.

2. Plan your program

Before you type it: Fix draw_polygon(sides, size) so it can draw ANY shape: loop sides times, and turn 360 / sides at each corner. The last line calls draw_polygon(6, 80), which should draw a closed hexagon. Write the steps in plain English (pseudocode), then the Python you will use.

3. Predict the drawing: three turns

What shape does the turtle draw? Code: import turtle ↵ t = turtle.Turtle() ↵ for i in range(3): ↵ t.forward(100) ↵ t.left(120)

a. A. A triangle
b. B. A square
c. C. A hexagon
d. D. A straight line

4. Match each shape to the turn at every corner

Draw a line from each word to its match.

a. Triangle ..... A. 45 degrees
b. Square ..... B. 60 degrees
c. Pentagon ..... C. 72 degrees
d. Hexagon ..... D. 90 degrees
e. Octagon ..... E. 120 degrees

5. Plan your program

Before you type it: Debug it! This program should draw a closed orange square, but it draws one long line. Fix the two bugs. Write the steps in plain English (pseudocode), then the Python you will use.

How did it go? Colour a face: 😀 I can do it · 🙂 Nearly there · 🤔 I need help

☐ I can move and turn the turtle to draw lines.
☐ I can work out the turn angle for any shape (360 ÷ sides).
☐ I can write a function with parameters that draws a shape.
☐ I can repeat and rotate a shape to make a pattern.